Gradient Descent:-

Gradient Descent is one of the most popular algorithms to perform optimization and by far the most common way to optimize neural networks.

Type of Gradient Descent
- Batch
- Stochastic
- Mini batch.

① Batch GD

In Batch GD the value is update only no. of times of epochs.

② Stochastic GD

epoch → 10

for i in range (10):
    shuffle
    for i in range (X.Shape[0]) → 80 rows
        → 1 random point
        → y_hat → forward se nikal lengy
        → loss
        → update w, b → $W_n = W_n - \eta \frac{\partial L}{\partial w}$

    avg loss print

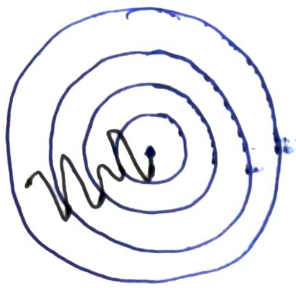
no. of update = epoch × no. of Data (rows)

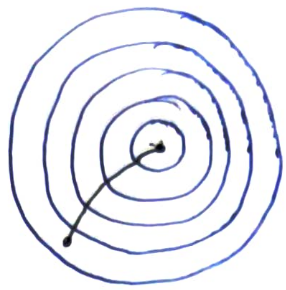
frequency of weight update is higher.

Ques: Which is faster (given same no. of epochs)
↳ Batch is faster.

Ques: Which is the faster to converge (given same epochs)
↳ Stochastic GD.

loss decrease in stochastic in unstable manner.
and Batch → stable manner.

Stochastic 

Gradient 

SGD help the algo to move out of local minima.

SGD give approximate solution not exact.

③ Mini Batch GD

for i in epochs:
  for j in no. of batch
    1 batch
      ↳ y_pred
      ↳ loss
      ↳ update

faster

Batch > Mini > SGD.

In converging

Batch < Mini < SGD

Here we update the parameters using small subset (batch).

Ques: why batch_size is provided in multiple of (2)?
↳ 2, 4, 8, 32, 64, ... for use the RAM effective. and make it faster.

Ques: What if batch_size doesn't divide # rows properly

eg:- rows n = 400
batch_size = 150.

400 → 150
400 → 150
400 → 100 (left rows).